

Cloud Computing Fundamentals

Learn complex cloud architecture using simple daily school situations.

The School Computer Exam Problem

? How can ALL students complete the exam WITHOUT BUYING 80 more computers? ?

SCENARIO:
100 STUDENTS,
ONLY 20 COMPUTERS!



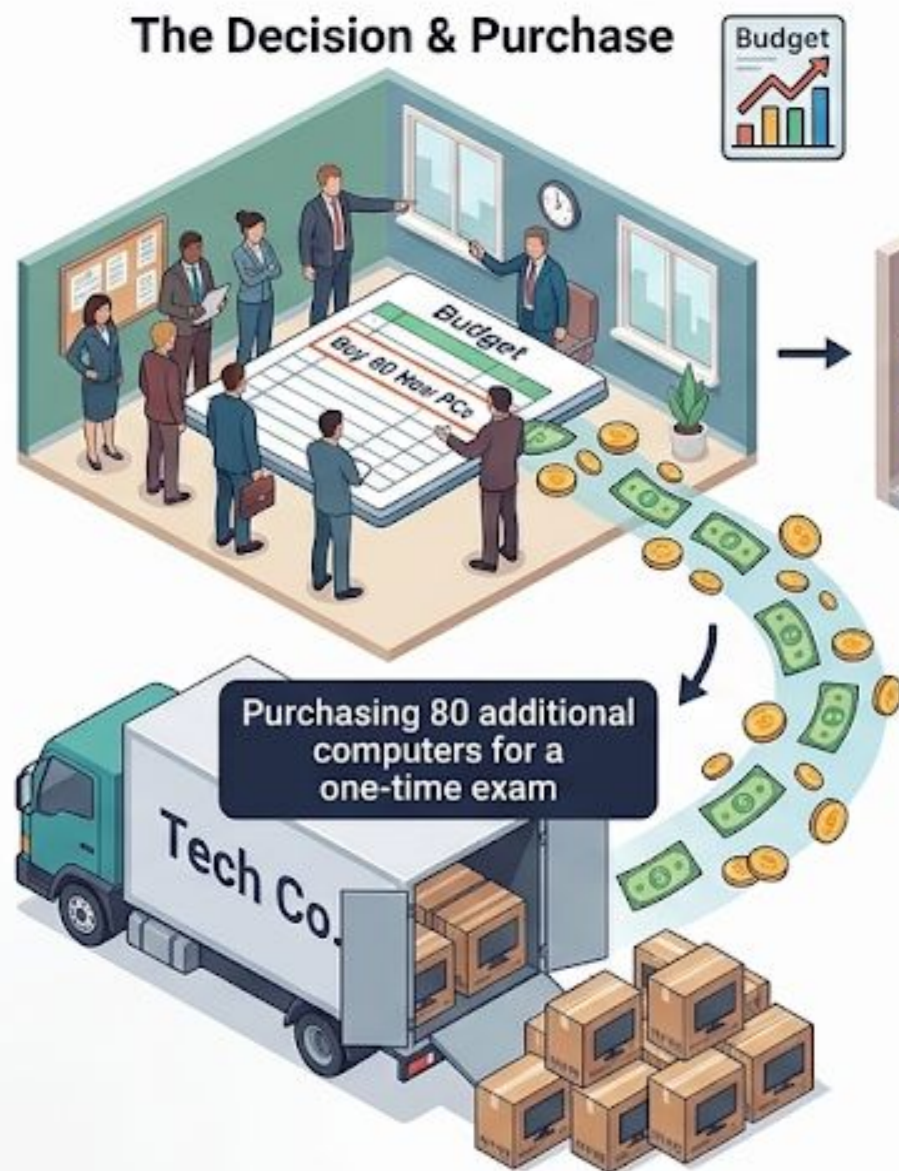
Students
Waiting



Which Cloud Solutions can help?
(e.g., VDI, Streaming Apps)

Traditional Solution: Overspending for Peak Demand

The Decision & Purchase



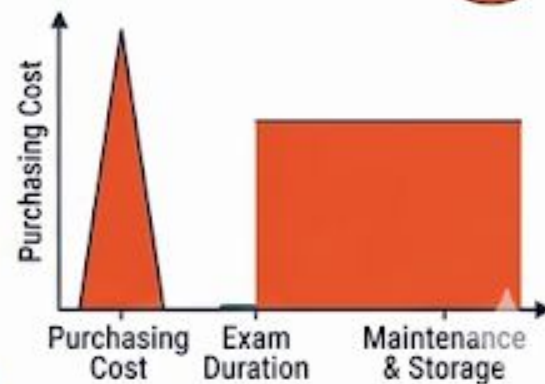
The Stacks of Goods



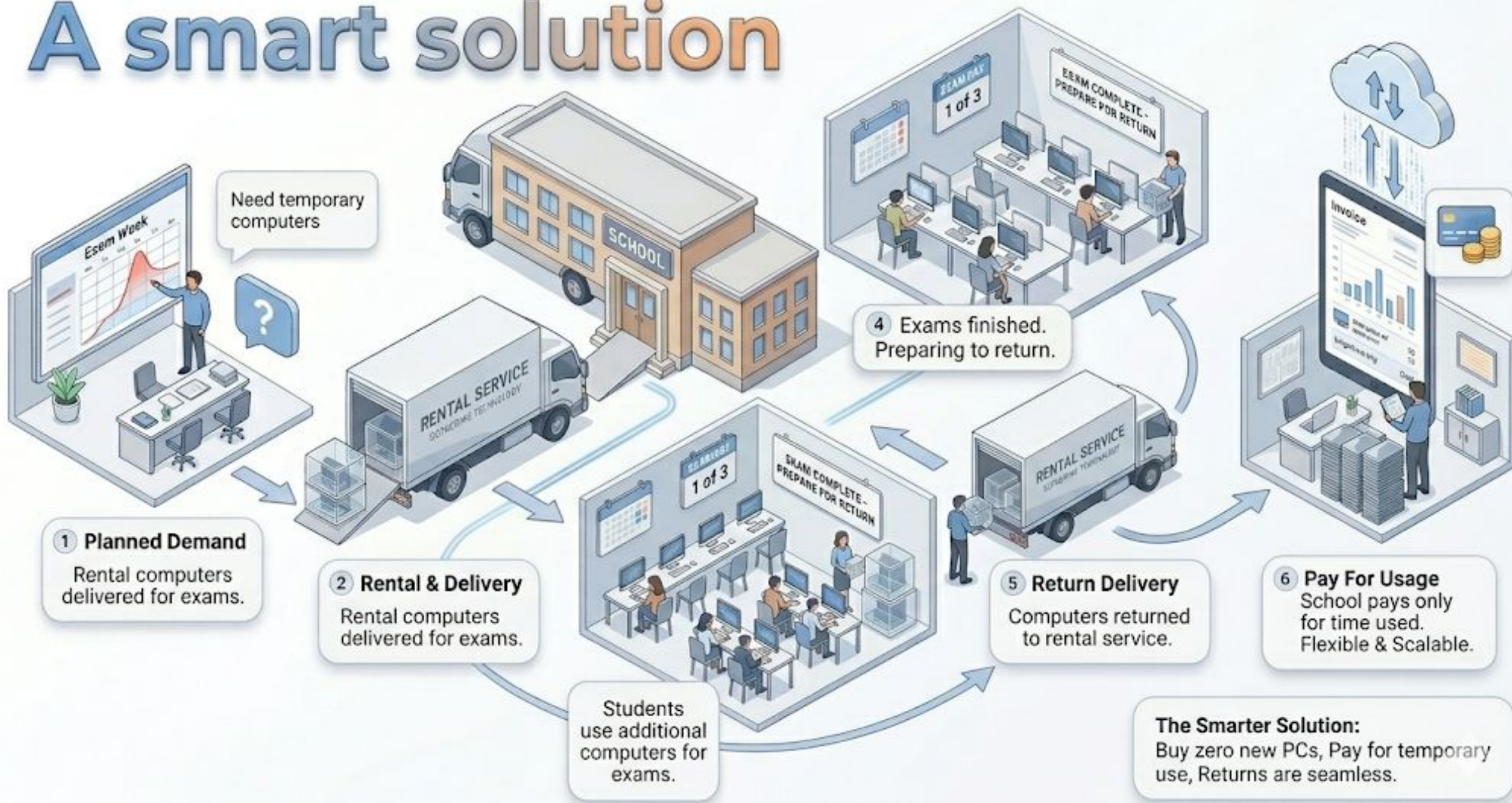
75% of Computers Idle Post-Exam
(80 of 100)



The Financial Bleed



A smart solution



WHAT IS CLOUD COMPUTING?

'THE SCHOOL COMPUTER LAB DILEMMA'



School uses 20 computers normally, but needs 100 for exams.

'CHOOSE YOUR OWN ADVENTURE'

OPTION A: BUY 80 PCs PERMANENTLY



High cost, storage issues, most go unused after exams.

OPTION B: RENT EXTRA PCs ONLY FOR EXAM



Temporary cost, efficient, returns after use.

'CLOUD COMPUTING IS LIKE RENTING COMPUTERS!'

Instantly access additional powerful computers and software from the internet, as a service!



benefits



Scalability:
Use more when needed, less when not.



Always Updated:
Get the latest tech without buying new hardware.

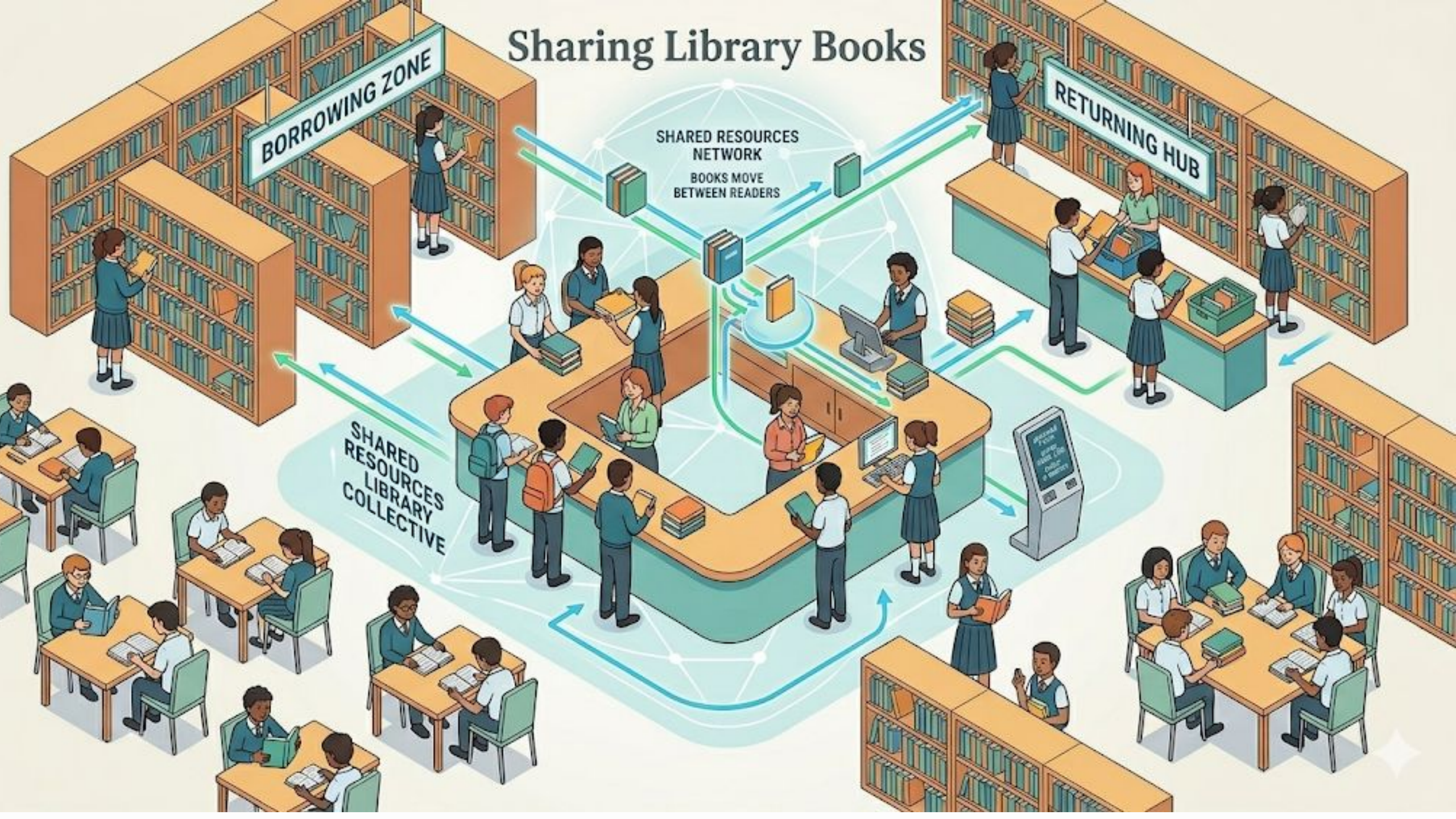


Accessible Anywhere:
Access your work from home or school.



It's a flexible solution. Cloud Computing delivers technology on-demand, like the school renting computers only for exams.

Sharing Library Books



Resource Pooling



Library

Books → Resources

Students → Users

Borrowing → Usage



Cloud Pool

Shared resources used by multiple users when needed.

SELF-SERVICE LIBRARY COUNTER



CLOUD COMPUTING FUNDAMENTALS: ON-DEMAND SELF-SERVICE

LIBRARY CONCEPT (ANALOGY)



2. CATALOG
(Book Selection)



3. REQUEST & AUTHORIZE
(Librarian Approval)



4. GET RESOURCE
(Take Home Book)

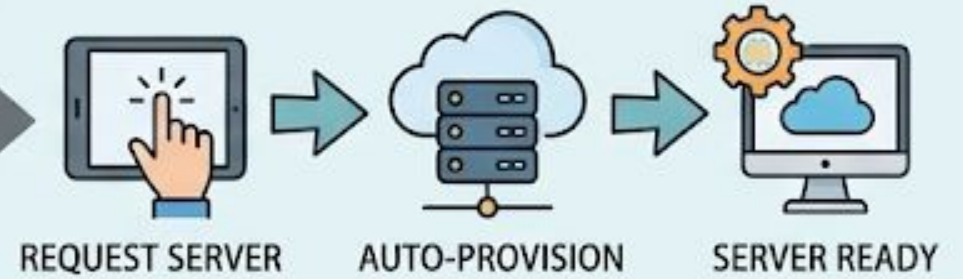
MAPPING



Student → Cloud User



CLOUD CONCEPT (FUNDAMENTALS)



1. CLOUD USER
(via Portal/API)



2. AUTO-PROVISIONING
(Compute, Storage)



3. DEPLOY & CONFIG
(Self-Service)



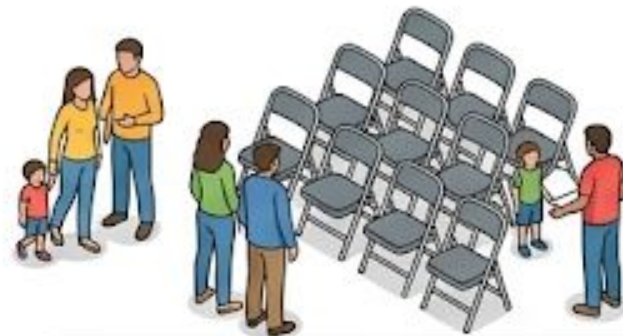
4. INSTANT ACCESS
(No human interaction)

THE CLOUD USER PROVISIONS RESOURCES DIRECTLY, ON-DEMAND, LIKE GETTING BOOKS WITHOUT A LONG LINE.

Sports Day Seating Problem

Scenario

The school is holding its annual sports day event, but many parents have arrived and there is a critical shortage of chairs for them to sit on, causing discomfort and disorganization.



Arriving Parents



Parents Arrived: **250+**



Current Seating Status



Chairs Available: **100**



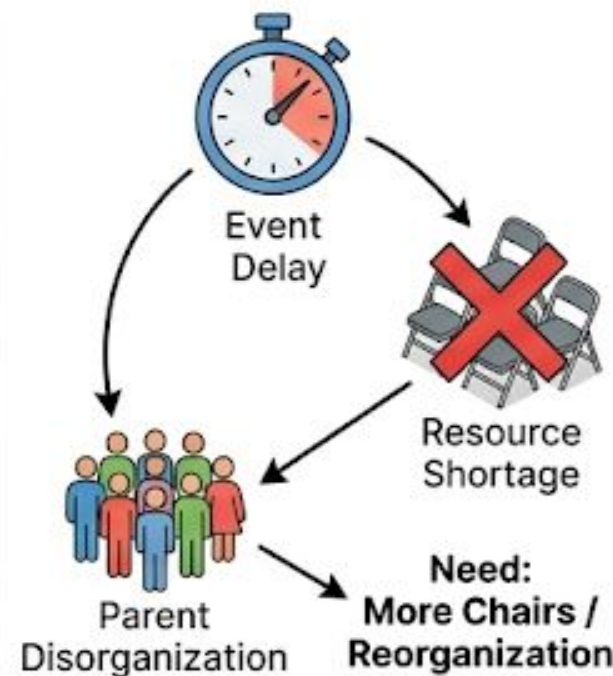
Problem

(Parent Experience)



STANDING-ROOM ONLY!

Impact & Preparation



Solution: The school must find a quick way to provide more seating, manage the flow, or communicate alternative arrangements. Efficient resource management is key.





RAPID ELASTICITY

NORMAL SCHOOL DAY:

100 students, 100 chairs,
everything works normally.



DEMAND GROWS

TEMPORARY ADDITION

SPORTS DAY:

1000 visitors arrive, the school temporarily adds many extra chairs.



DEMAND FALLS

REMOVED

AFTER SPORTS DAY:

Extra chairs removed and returned,
the school goes back to normal.



Cloud Mapping



Extra Chairs

→ Additional Cloud Resources



Adding Chairs

→ Scaling Up



Removing Chairs

→ Scaling Down

KEY MESSAGE: Cloud resources automatically increase when demand grows and decrease when demand falls.

HOW PAYING FOR FOOD WORKS IN THE SCHOOL CANTEEN: ONLY PAY FOR WHAT YOU CONSUME.

1 BROWSE AND SELECT YOUR FOOD



2 CHECKOUT AND ITEMIZATION



Your bill lists only your choices.



agreement

charges avoided

3 PAYMENT FOR SELECTED ITEMS

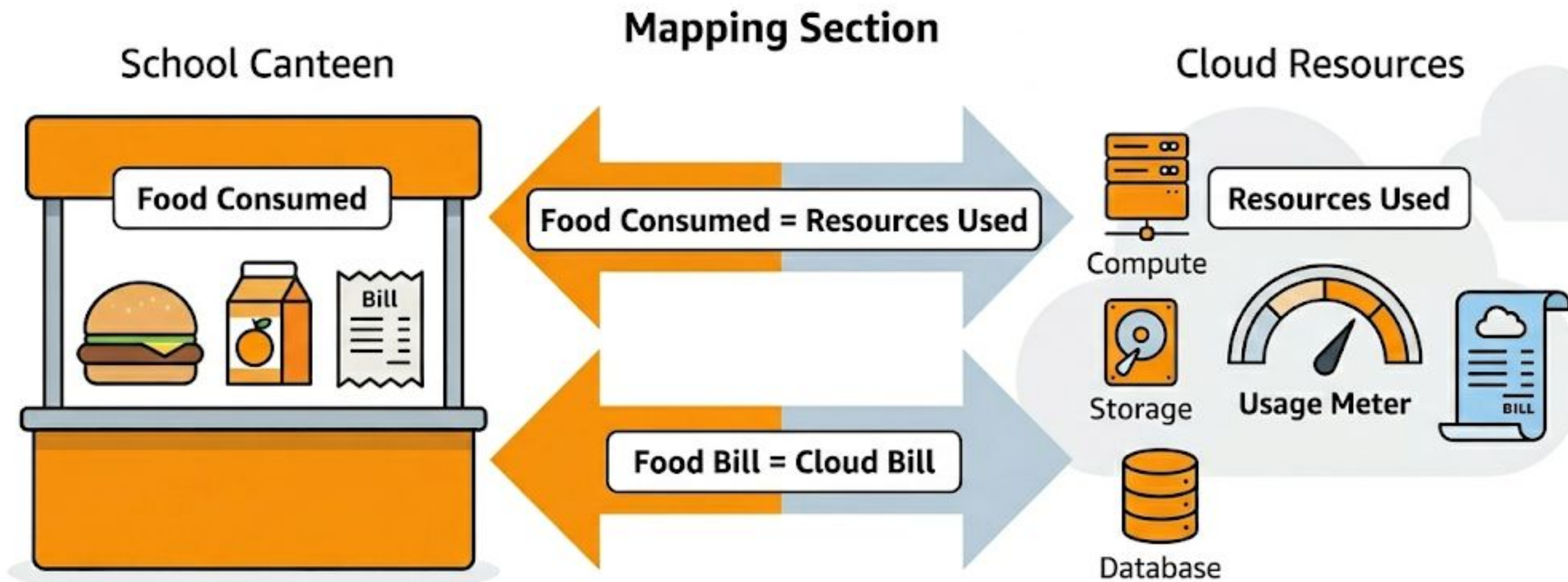


No Unpurchased Charges

Simple & Fair Billing

This is educational and helps students understand value and only pay for their food, not others or wasted items.

Measured Service (Pay Only For What You Use)

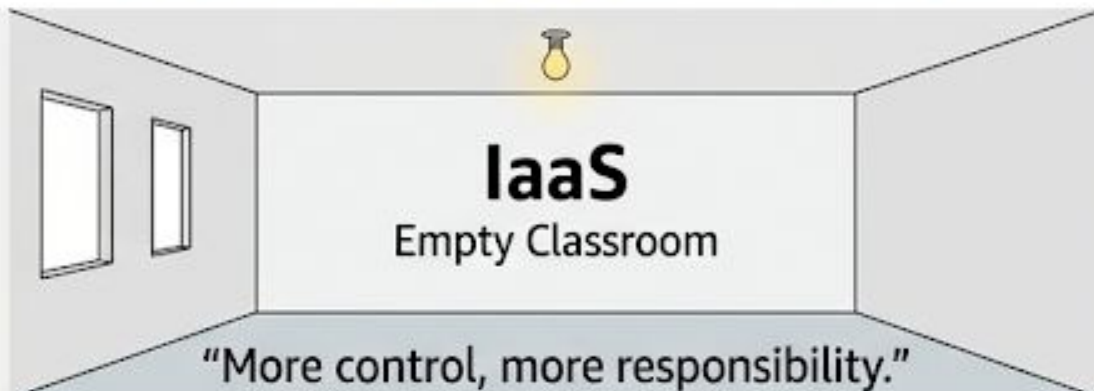


**Just as students pay only for the food they consume,
cloud users pay only for the resources they use.**

Cloud Service Models

Cloud Provider Gives:

- ☒ Classroom
- ☒ Electricity
- ☒ Basic Infrastructure

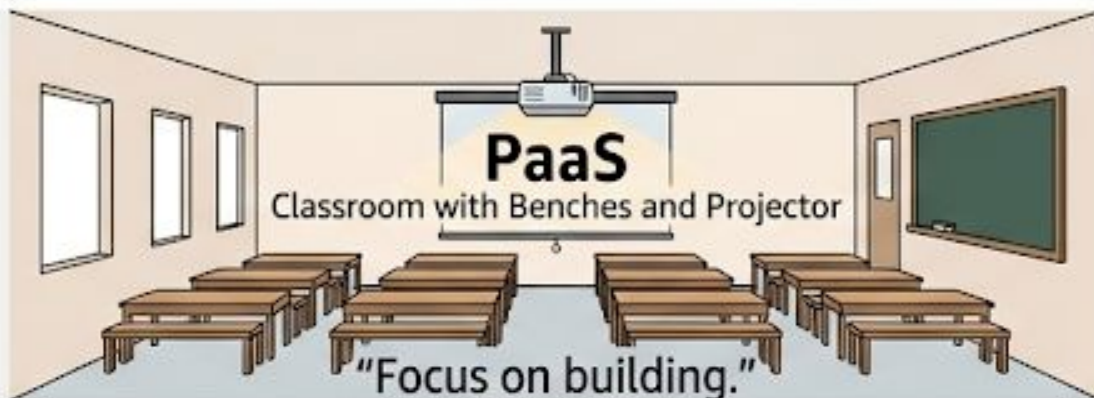


Student Must Arrange:

- ☒ Benches
- ☒ Books
- ☒ Teaching Materials

Cloud Provider Gives:

- ☒ Infrastructure
- ☒ Platform
- ☒ Tools



Student Focuses On:

- ☒ Creating projects
- ☒ Learning activities

Cloud Provider Gives:

- ☒ Classroom
- ☒ Teacher
- ☒ Books
- ☒ Projector



Student Only Uses:

- ☒ the Service

IaaS → Build Everything



PaaS → Build Application



SaaS → Just Use



Cloud Deployment Models

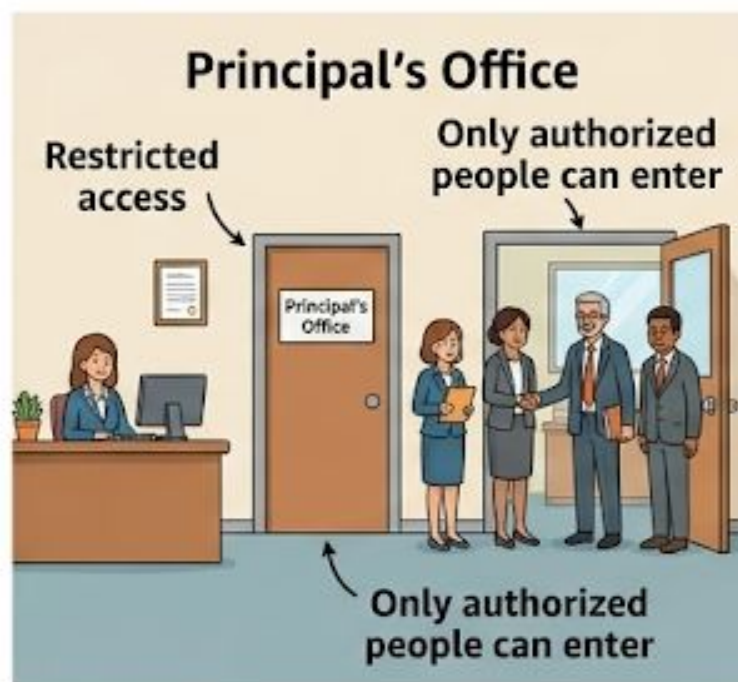


Public Cloud



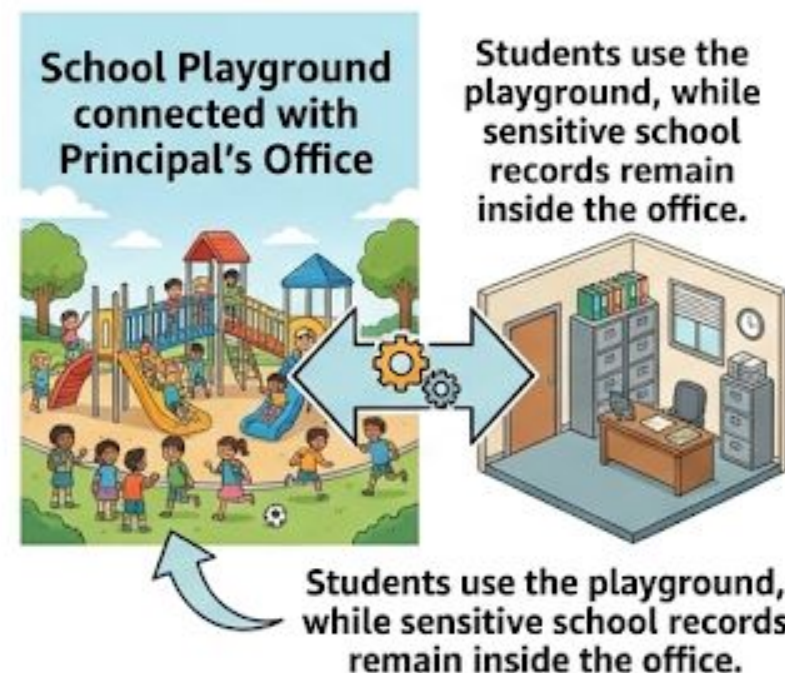
Shared by many users.

Private Cloud



Dedicated and controlled.

Hybrid Cloud



Combines both environments.



Public Cloud → Shared

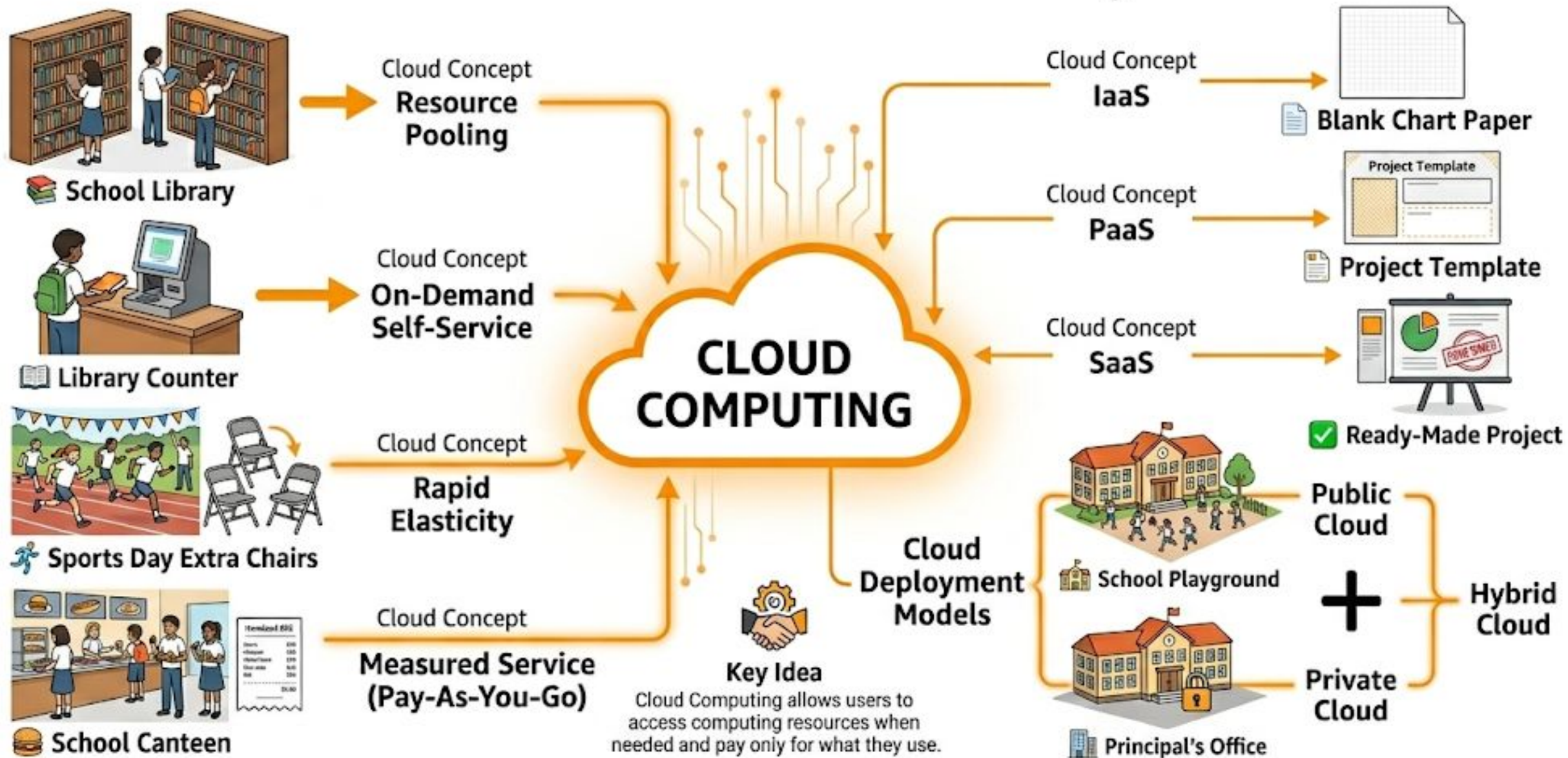


Private Cloud → Dedicated



Hybrid Cloud → Combination

What We Learned Today



Question: "If cloud resources are available everywhere, where are they actually located?" | **Next Topic:** "AWS Global Infrastructure"